

L01 + L02 — The Complete Week

Thermal equilibrium, from a cold brick to a law with the number zero

Four pages in one volume: **L01 Pre-class** (the cold brick and the micro↔macro bridge), **L01 Deep-dive** (the handwritten review — 8/8 concepts), **L02 Pre-class** (Conservation of Energy and the Zeroth Law — predictions preserved, answers withheld), and **L02 Deep-dive** (post-class resolutions: the Noether twist, the IR-demo sequence, and the typed review resolved 9/9 after feedback). Figures are print-stills of the interactive animations.

Instructor Dr. Simin Najmi · Section 102 | **Lectures** Sept 9 & Sept 11, 2026

Interactive edition finneaswang.github.io/phys-course-os | **Vault** Obsidian · PHYS 157

Sources L1/L2 pre- & post-class slides (post-class canonical) · calibrated lecture transcripts · the student's handwritten and typed reviews

QA review copy — physics accuracy, lecture-boundary discipline, grading fairness. A4, vector text.

The cold brick — and the two pillars

L01 PRE-CLASS · STEP 1 / 10

How this works, and the two pillars

Ten short steps, each deeper than the last. **By the end you can:** tell the cold-brick story twice (macro & micro); distinguish temperature, heat, and thermal equilibrium; explain why thermodynamics is possible at all; and say what L01 deliberately does *not* contain.

Pillar 1 — Thermodynamics (first half): macroscopic variables (T , V , p), thermal expansion, phase change, heat and its transfer methods, equilibrium. Built on averages over $\sim 10^{23}$ particles.

Pillar 2 — Oscillations & Waves (second half): *collective* motion of the same molecules — guitar strings, sound, ocean waves. Described by a wave equation; again no per-particle tracking.

⚠ L01 contains **no formulas**. If you catch yourself deriving one here, you're ahead of the lecture — park it.

L01 PRE-CLASS · STEP 2 / 10

The cold brick — predict first

September 9, Hebb building. A brick out of a cooler. First guess: *liquid nitrogen*. Wrong — **dry ice**, so it starts at about $-78.5\text{ }^{\circ}\text{C}$, then sits in the $\sim 20\text{ }^{\circ}\text{C}$ room.

Predict, then check against the slides — *in the first minute, what do you actually see?*

(source + guided physical inference: the photos show what happened; the why is labelled inference, not lecture content)

✗ A · Nothing — unchanged. Look at the temperature difference: $-78.5\text{ }^{\circ}\text{C}$ vs $\sim 20\text{ }^{\circ}\text{C}$. Something dramatic happens at the surface.

✓ B · Frost forms on the brick's surface. Water vapour in the air touches a surface far below $0\text{ }^{\circ}\text{C}$ and leaves the gas phase → frost forms almost immediately — exactly what the ~ 15 min photo shows. Heat starts flowing in and the brick begins warming.

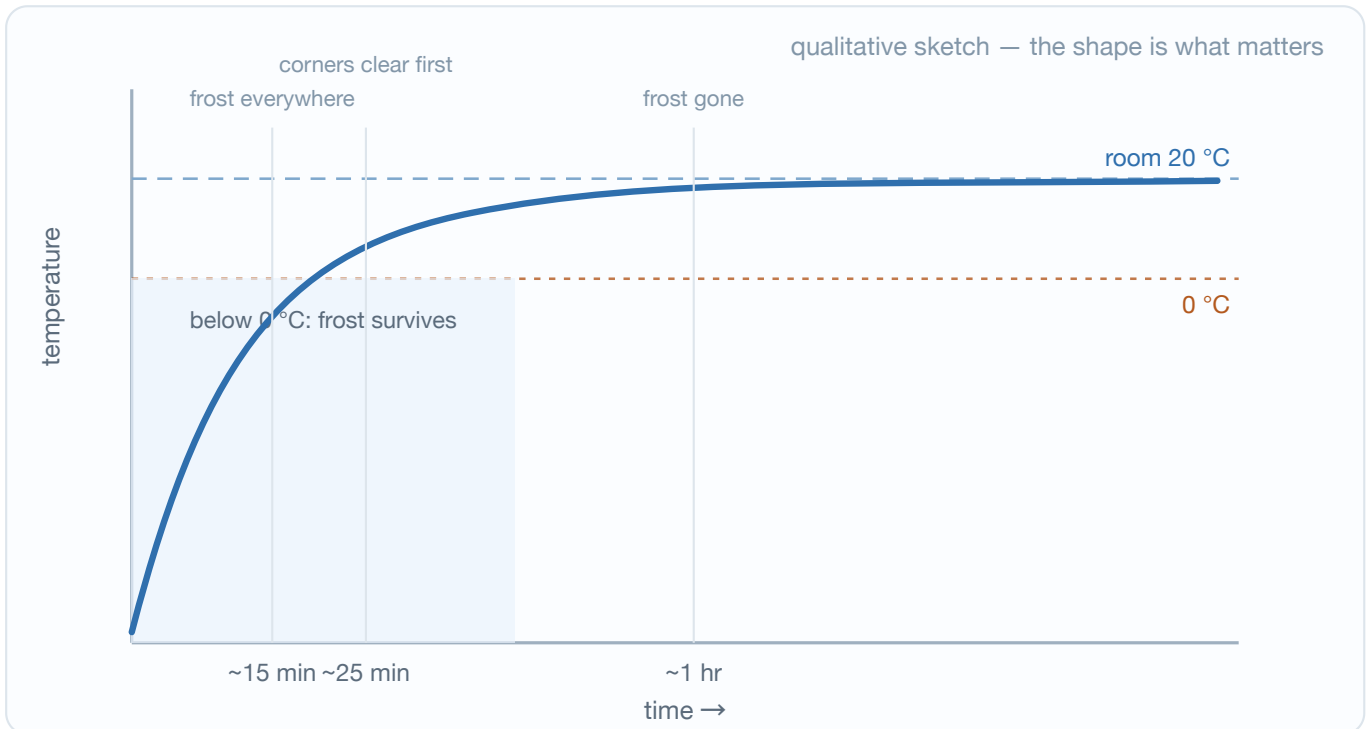
✗ C · The brick glows faintly. Glowing usually means visible light — that needs a far higher temperature. The brick IS emitting thermal radiation right now, mainly in the infrared, which your eyes cannot detect.

✗ D · Water pools underneath it. Meltwater only appears once the surface climbs past $0\text{ }^{\circ}\text{C}$ — and the ~ 1 hr photo shows the frost gone, not water pooling. First comes frost.

What actually happened (slide photos)

- **~ 15 min:** frost over the whole brick.

- **~25 min:** frost retreating — **corners and edges clear first** (most exposed surface per amount of brick → they warm fastest).
- **~1 hr:** brick nearly clear, temperature approaching room temperature.



The brick's temperature over its hour: it approaches room temperature — fast at first, then slower. Frost survives only below the 0 °C line. **Fast-then-slow is a qualitative preview — not established in L01.**

Wording trap: slides say *condensation*, but at -78.5 °C the vapour goes from gas **straight to solid** — deposition (凝华). For the quiz, answer with the instructor's wording; keep "deposition" as private precision.

L01 PRE-CLASS · STEP 3 / 10

The macroscopic view

Macroscopic = the scale you can see or measure. Five measurable statements:

- **Heat flowed in** — from everything warmer. (*Faster-with-bigger- ΔT is a preview inference — L01 shows the flow, not the rate law.*)
- **Its temperature rose** — continuously, then more slowly.
- **Its dimensions grew** — every linear dimension, slightly (thermal expansion — first real topic).
- **Frost formed, then cleared** — a phase change in the air next to the brick.
- **It stopped changing** — brick and room at the same temperature: thermal equilibrium.

This language never mentions atoms. That's the point — macroscopic variables are chosen so you *don't need* them.

L01 PRE-CLASS · STEP 4 / 10

The microscopic view

- **The brick's atoms vibrate** about fixed equilibrium positions (solid — contrast with gas molecules, which travel). Cold brick = slow, low-energy vibrations.
- **Air molecules collide with it** — each impact transfers a bit of momentum and energy; billions per second = "heat flowing in".
- **Everything radiates** — wavelength set by temperature; room temperature → mainly infrared, invisible. IR-camera demo promised.

Go deeper: individual collisions can transfer energy in either direction. Because the warmer side transfers more on average, there is a **net** flow from hot to cold. The word doing the work is **net**.

L01 PRE-CLASS · STEP 5 / 10

The bridge: averages become variables

microscopic behaviour → averages → macroscopic variables → thermodynamics

- The brick's **size** (Volume) is **related to** the average spacing between atoms — why expansion is visible at all.
- The brick's **temperature** ↔ the **average kinetic energy** of its particles. Prof's phrasing: "*the average of all those kinetic-energy values is what we call temperature.*"

That is **Thermodynamics**: summarizing 10^{23} things into a handful of variables, then finding the rules those summaries obey.

L01 PRE-CLASS · STEP 6 / 10

Temperature vs Heat vs Equilibrium

Temperature — a **property of the object**, a *state variable* (状态量). In the L01 model: the average kinetic energy of its particles.

Heat — **energy in transit**, driven by a temperature difference; exists only *while crossing a boundary*; carried by collisions and radiation.

Thermal Equilibrium — endpoint of heat flow: common temperature; the **net** exchange drops to zero because the averages have matched. (*Microscopic interpretation.*)

The two traps: ① "the brick contains heat" — wrong: it contains internal energy; heat is energy *while moving between things*. ② "at equilibrium everything stops" — wrong: only the **net** flow stops; collisions continue both ways. Equilibrium = balanced traffic, not an empty road.

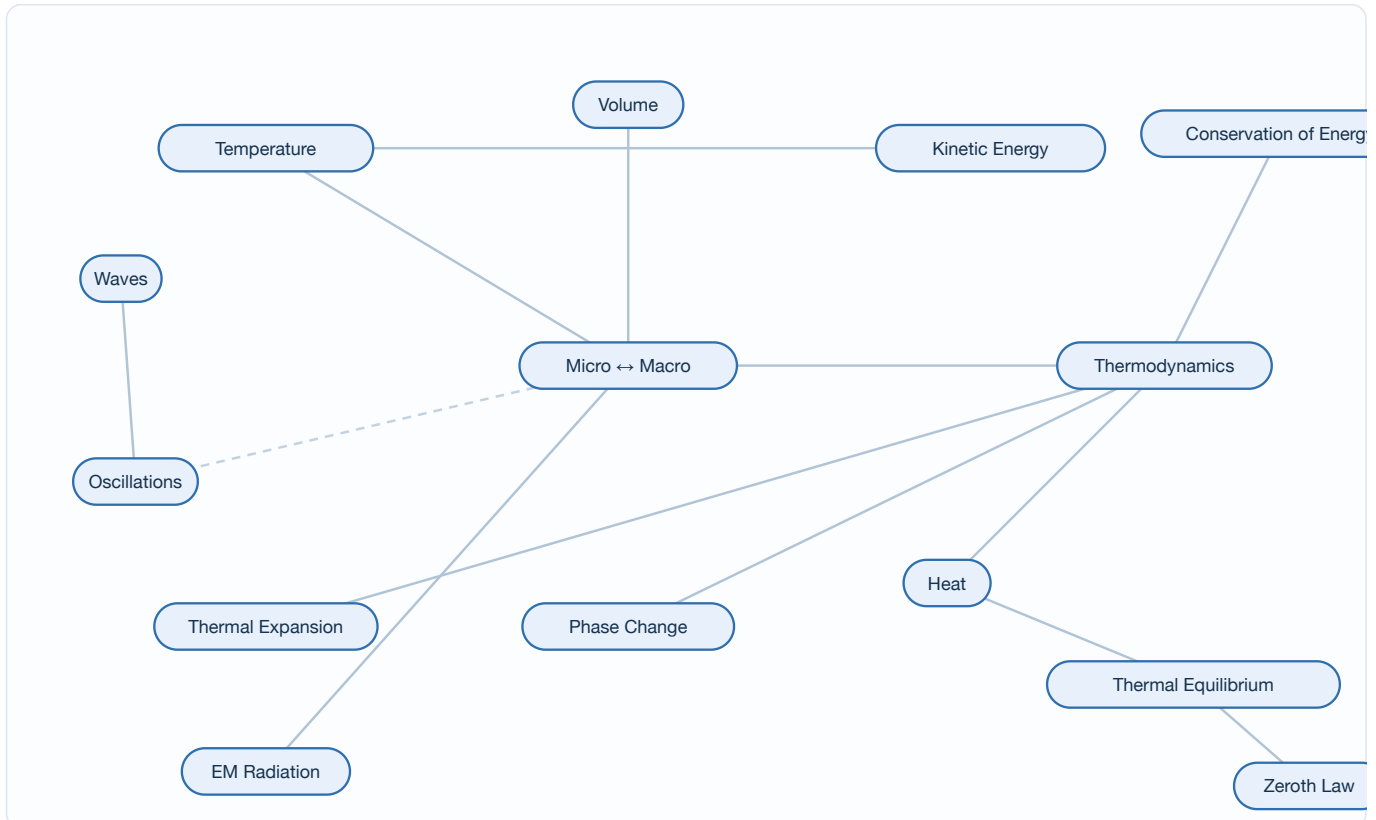
L01 PRE-CLASS · STEP 7 / 10

The waves teaser

- The brick is also hit by **sound** and light — **waves**: collective motion, **energy travels without the material travelling**.
- Sound waves, guitar strings, ocean waves; transverse and longitudinal oscillations.
- Same trick as thermodynamics: no per-particle tracking — a **wave equation** summarizes it.

L01 PRE-CLASS · STEP 8 / 10

Your concept graph



14 concept pages after L02 — dashed = previewed, not yet taught.

L01 PRE-CLASS · STEP 9 / 10

Check yourself

1 · Why is tracking every particle impossible — and why is that fine?

~ 10^{23} particles; the **averages** are all that matter, and averages appear as macroscopic variables. Summaries + rules = thermodynamics.

2 · How does a brick atom's motion differ from an air molecule's?

Brick atoms **vibrate about fixed positions**; gas molecules **travel long distances** between collisions.

3 · Which is the state variable — temperature or heat?

Temperature. Heat is **energy in transit**, driven by ΔT , carried by collisions and radiation.

4 · At thermal equilibrium, what exactly stops — and what continues?

The **net** flow stops; individual two-way exchanges continue forever.

5 · Why did the frost clear from corners and edges first?

Most **exposed surface per amount of brick** → fastest warming. Geometry controls heat-flow rate.

6 · The brick radiates constantly — why can't you see it?

Room-temperature emission is mainly **infrared** — outside the narrow visible band our eyes detect.

L01 PRE-CLASS · STEP 10 / 10

60-second review

- Macroscopic variables = **averages** over $\sim 10^{23}$ particles.
- **Temperature** = average kinetic energy *in the L01 model* — a state variable.
- **Heat** = energy in transit (collisions + radiation); never "stored".
- Cold brick: heat in → warms → expands; frost **deposits**, clears **corners first**.
- **Equilibrium**: same T, net flow zero, two-way collisions continue.
- Everything radiates; room-T = infrared, invisible. Waves = collective motion.

Your handwritten review, graded — 8/8 concepts

Verdict: 8/8 concepts are recoverable — but not 8/8 written-response quality. One answer needs wording repair (Q2), two were sharpened, and the live practice session surfaced two bigger gaps — they lead the re-test ledger.

#	Question	Verdict & the chain that should be written
1	Macroscopic picture of the brick?	✓ correct. warmer surroundings → heat flows into brick → temperature increases → brick expands → thermal equilibrium. <i>Also macroscopic: the frost (deposition).</i>
2	Microscopic picture?	✓ concept ok · wording repair. air molecules collide with the brick → energy is transferred → atoms vibrate with higher energy. <i>Draft read "vibrate with higher energy is transferred to the brick" — intent right, sentence doesn't parse. Chain order matters on written questions.</i>
3	Track every particle?	✓ correct. too many particles → use averages → describe the whole system. <i>Memorize verbatim.</i>
4	Air molecules doing?	✓ correct. colliding with the brick, transferring energy to it.
5	Lasting till...?	✓ sharpened. net transfer continues until thermal equilibrium — net flow 0, collisions continue both ways .
6	What determines macroscopic size?	✓ correct. the average spacing between its atoms.
7	What does temperature represent?	✓ correct. the average kinetic energy of its particles (L01 model framing).
8	Must an object glow to radiate?	✓ sharpened. All objects emit EM radiation, regardless of temperature; temperature sets wavelength/frequency — ordinary temperatures → mainly infrared. (Draft's "depending on their temperature" reads as "only if hot enough".)

Worked example — the two-brick question

Brick A at 20 °C, **Brick B** at 100 °C, same room. (a) Do both emit EM radiation? (b) Would the radiation from A and B have exactly the same characteristics? (c) Direction of **net** heat flow for each? (d) As B cools?

(a)

Yes — both. Every object emits regardless of temperature.

(b)

No. Different temperatures ⇒ different characteristic wavelength/frequency. (L01 stops here — it does not ask which pours out more energy.)

(c)

A: in equilibrium with the room $\rightarrow$ net flow = 0 (two-way exchanges cancel). **B:** hotter than the room $\rightarrow$ net flow from B into the air.

(d)

ΔT shrinks $\rightarrow$ net flow shrinks $\rightarrow$ **zero** once B eventually reaches the same temperature as the room — thermal equilibrium, in the instructor's own words.

Variations

V1 · A very large cool metal block and a tiny hot metal bead touch. Which way does heat flow? Does the larger object's greater total internal energy decide it?

Heat flows from the **hotter bead to the cooler block**. Direction is set by **temperature difference**, not by size or total energy.

V2 · At the instant of equilibrium, what is zero and what is not?

Zero: net energy flow. **Not zero:** the individual two-way exchanges — balanced traffic, not an empty road.

V3 · The frost is gone by ~1 hr. What does L01 let you say?

Frost forms on the very cold brick and later disappears as the brick warms. That is the full claim.
(Vocabulary: slides say "condensation"; strictly, gas $\rightarrow$ solid frost is closer to deposition.)

Re-test ledger — prioritized from live practice

- ① **Macro vs Micro recognition** — when asked a macroscopic question, the first instinct reached for particle motion.
- ② **Macro $\leftrightarrow$ Micro mappings** — size $\leftrightarrow$ **average atomic spacing** (first answered "temperature"); temperature $\leftrightarrow$ **average kinetic energy**.
- ③ **Thermal equilibrium** — net flow = 0; exchanges continue both ways.
- ④ **Universal radiation statement.**
- ⑤ **Written-answer chain order** — cause $\rightarrow$ effect, one arrow per step.

The law under the demo

L02 PRE-CLASS · STEP 1 / 8

Conservation of Energy

The law, as the slides state it:

- Each part of a physical system has a certain amount of energy.
- The total energy of an **isolated** system doesn't change with time.
- BUT energy can be **transferred between different parts** and take **different forms**.
- In thermodynamics: the **microscopic kinetic and potential energy of atoms and molecules**.

Keep precise: **isolated** (nothing enters or leaves — total locked) and **transfer** (energy moves between parts and changes form — that's what heat is).

Clicker 1 — *Conservation of energy was first fully understood by...* A · Newton · B · Einstein · C · Emmy Noether · D · James Joule. **Answer deliberately not revealed on a pre-class page** — commit to a guess; the instructor resolves it in the lecture.

Connect it back to the brick: L01's "heat flows until equilibrium" **silently used this law** — what leaves the air is exactly what the brick gains.

L02 PRE-CLASS · STEP 2 / 8

Heating at the molecular level

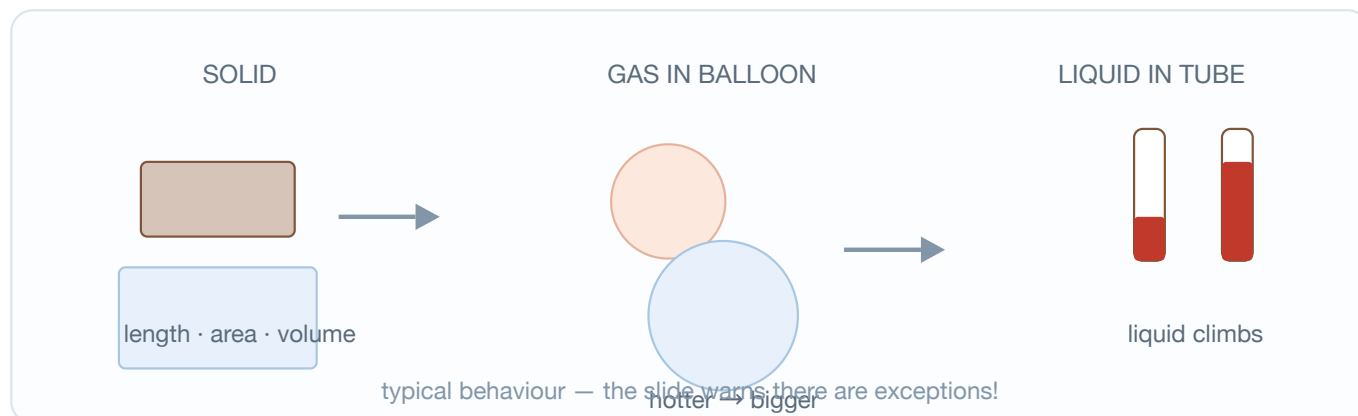
- **Heating = adding energy at the molecular level**; cooling = removing it.
- The hot brick's atoms carry **more kinetic AND more potential energy** than the cold brick's.

Lecture extension: why does a vibration hold potential energy? A vibrating atom pushes against its neighbours — the stored push is that potential energy.

L02 PRE-CLASS · STEP 3 / 8

Macroscopic properties

- **Most of them change** — often only slightly for small ΔT : **size, volume, density; IR output** at different frequencies; e.g. **electrical conductivity**.
- **Thermal expansion — typical, with exceptions:** solids, gases in balloons, liquids in tubes: hotter → bigger. "Size" = **length, area, or volume**.
- Hotter → **more IR radiation**.



Expansion trio: the solid grows in every dimension, the balloon's gas expands, the tube's liquid climbs — "size" covers length, area, and volume.

L02 PRE-CLASS · STEP 4 / 8

Clicker 2 — predict: contact between two blocks

What happens if we put a hot block in contact with a room-temperature block?

- A · Nothing B · Heat flows until they are the same temperature C · The hot block cools to room temperature
D · The room-temperature block heats to the hot block's temperature E · None of the above

Answer deliberately not revealed on a pre-class page — the lecture resolves it (see Part IV).

L02 PRE-CLASS · STEP 5 / 8

The rule of contact: three outcomes

1 · **Nothing changes** → the systems are in **EQUILIBRIUM, SAME TEMPERATURE.**

2 · **Energy flows brick** → **balloon** = flow of HEAT; brick has the higher temperature.

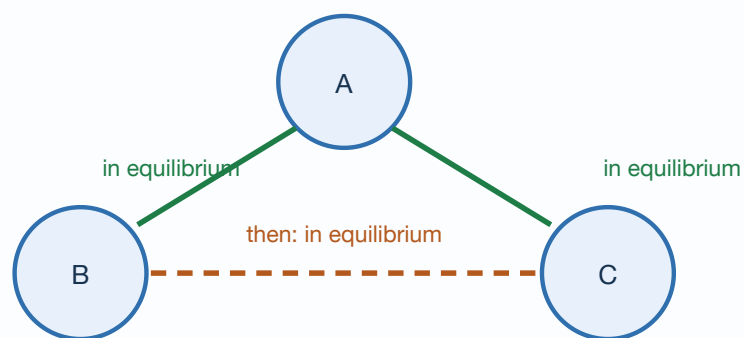
3 · **Same, other direction** — balloon hotter.

Course wording confirmed: **heat = thermal energy flowing from one object to another because of a temperature difference.** Direction set by temperature — never by "who has more energy".

L02 PRE-CLASS · STEP 6 / 8

The Zeroth Law — why thermometers work

If A is in equilibrium with B, and A is in equilibrium with C, then B and C are in equilibrium with each other. "Otherwise, temperature wouldn't make sense!"



$A-B \checkmark$ and $A-C \checkmark \Rightarrow B-C \checkmark$ — otherwise, temperature wouldn't make sense

The Zeroth Law triangle: two confirmed equilibria force the third.

Clicker 3 — A mercury thermometer sits in a glass of water and reads 20°C . We can conclude: A · water is 20°C · B · mercury is 20°C · C · both · D · neither. **Answer deliberately not revealed on a pre-class page** — the post-class deep-dive unpacks it (Part IV).

L02 PRE-CLASS · STEP 7 / 8

Demo prediction: the IR camera

Two aluminium blocks — one off the hot plate, one left in the room — placed in contact and watched with an **IR camera** (hotter = brighter). Sketch what you expect:

- (a) just as the blocks touch;
- (b) after a short time;
- (c) after a long time.

No predicted answers on a pre-class page — the activity is preserved for lecture. Hints that ARE on the slides: the camera displays hotter objects as brighter, and it reads infrared emission (the "every object radiates" physics from L01).

L02 PRE-CLASS · STEP 8 / 8

60-second review

- **Conservation of Energy:** isolated system's total constant; energy transfers between parts and changes form; thermo tracks microscopic KE **and** PE.
- Heating/cooling = adding/removing molecular energy; hot brick = more kinetic **and** potential energy.
- Most macroscopic properties shift with temperature — usually slightly. Expansion typical, **exceptions exist**.
- Contact = three outcomes; direction set by temperature, never by total energy.
- **Zeroth Law:** $A-B$ and $A-C$ in equilibrium $\Rightarrow B-C$ in equilibrium. Thermometers work because of it.
- Next class: **methods for measuring temperature**.

Post-class answers + the typed review, graded

Clicker resolutions: C1 → **C · Emmy Noether** (not Joule — the trap!) · C2 → **B** · C3 → **official key unresolved** in the available materials. **Typed review — after feedback: 9/9 concepts resolved** (initial gaps preserved in the table below).

A · Isaac Newton  21%

B · Albert Einstein  1%

C · Emmy Noether  ✓ **correct** · 19%

D · James Joule  **the trap** · 60%

class vote on Clicker 1 — the obvious answer was the trap

Why the unit of energy is named after Joule — and why that made him the wrong answer. Noether derived conservation from time symmetry; the lecture told her story (unpaid teaching in Germany, forced to move, finally published abroad).

#	Item	Verdict
1	Clicker 2 → B	✓ correct. "heat is not 'stored stuff'; heat is energy transfer due to temperature difference" — exactly the course definition.
2	Conservation of energy, own words	⚡ recovered after feedback. Initial answer conflated conservation of energy with thermal equilibrium. Final: "cannot be created or destroyed; only transferred or transformed" — the lecture's phrasing almost verbatim.
3	Particle energy types	⚡ recovered after feedback. Initial answer: kinetic energy only. Final: both kinetic and potential energy.
4	Why potential energy?	⚡ recovered after feedback. Initial answer: "everything has energy". Final: particles interact with surrounding particles; PE depends on distance . <i>Lecture adds: the interactions are electrical bonds; oscillation changes distance → PE changes.</i>
5	Add heat → internal energy?	✓ correct. it increases — and the lecture's new term internal energy (the microscopic KE + PE account) was used correctly.
6	Macroscopic properties	✓ correct. size, volume, density, radiation, conductivity — all five on the slides.
7	Hotter object's IR?	✓✓ sharpest answer in the set. stronger intensity and shorter wavelength — the lecture's exact two-part answer. Two-part questions need two-part answers.
8	Thermometer clicker (B/C — official key unresolved)	⚠ unresolved. Directly, the reading establishes the mercury's temperature. If thermal equilibrium with the water is assumed, the Zeroth Law implies the water matches — the intended key would then be C. The student's instinct moved between B and C during practice; both encode the same physics, and the assumption is the hinge.

- 9 Zeroth Law: A–B, B–C in equilibrium → A and C?
✓ **correct.** A and C in thermal equilibrium, same temperature (B correctly used as the bridge).

Worked example — the IR demo, resolved (slide 15)

Stage 1 • before contact

Each block in equilibrium *with something else*: the dark one with the room, the yellow one with the hot plate. Equilibrium is always with respect to something.

Stage 2 • thermal contact

The two-block system is now **isolated** (ignore table and air). Inside it: a temperature difference exists → heat must flow.

Stage 3 • during the flow

The slide's subtle line: the blocks are **not in equilibrium with each other, and each block is not in an equilibrium STATE** — some parts are hotter. Equilibrium requires temperature to be **uniform inside** each object too. This upgrades the ledger's precision item.

Stage 4 • the endpoint

Same temperature AND uniform — net flow zero, microscopic exchanges continue. One uniform brightness on the IR camera.

Variations — asked in the live demo

V1 • Does touching a thermometer lower your body temperature?

Any measurement **affects the system** — that's the rule. Here the effect is **negligible**: the thermometer is tiny compared to you.

V2 • Why did the eyes of people wearing glasses look dark on the IR camera?

Glass **does not transmit infrared**. The eyes radiate like the rest of the face — but their IR can't reach the camera through the lenses. Reading an instrument means knowing what it cannot see.

V3 • Three lamps (LED, CFL, incandescent) all look white — which is the incandescent on the IR photo?

The **brightest** one: incandescent bulbs waste most electric energy as **heat** (~5% becomes light) → huge IR output. CFL better; LED most efficient.

L2 Deep Phrases — trigger words

If you see...

Immediately think...

" isolated system"	nothing enters or leaves $\Rightarrow$ total energy constant
" internal energy "	the microscopic KE + PE account of the object
" add heat "	internal energy increases (transfer — not creation)
" why potential energy? "	interactions (electrical bonds); PE depends on distance , which oscillates
"hotter object's IR? "	intensity $\uparrow$ (brighter) AND wavelength shifts shorter
" direction of heat flow?"	hotter $\rightarrow$ colder; the WHY comes later (entropy)
" thermometer reads X "	that's the thermometer's temperature; the object's follows via equilibrium
" uniform temperature"	equilibrium inside the object too — not just between objects

Ledger update

- ① **Macro vs Micro recognition** — improved (kept separate throughout L2 answers).
- ② **Macro $\leftrightarrow$ Micro mappings** — improved (both used correctly).
- ③ **Equilibrium precision** — upgraded: equilibrium = same temperature **and uniform within** each object (slide 15).
- ④ **Universal radiation** — passed and extended: hotter $\rightarrow$ intensity $\uparrow$, wavelength shorter.
- ⑤ **Chain order** — holding.
- ⑥ **NEW — Isolated system & internal energy:** L02 vocabulary; expect it in every L03 question.
- ⑦ **NEW — History/fact traps:** Noether vs Joule — don't answer on autopilot from "the unit is named after him".

Every blue term in the interactive version opens one of these

INTRODUCED · L01 Microscopic & Macroscopic Description 微观与宏观描述

Two vocabularies for the same object: microscopic (vibrate about fixed sites in solids; translate and collide in gases; radiate) vs macroscopic (the few measurable averages). Chain: microscopic behaviour → averages → macroscopic variables → thermodynamic laws. Mappings: size ↔ spacing; $T \leftrightarrow \text{avg KE}$.

INTRODUCED · L01 Temperature 温度

A macroscopic variable and a **state variable**: in the model introduced in L01, temperature reflects the **average kinetic energy** of a system's particles. Sets the direction of spontaneous heat flow and the endpoint (equilibrium).

INTRODUCED · L01 Volume 体积

Macroscopic variable **related to the average particle spacing** (for a given amount and structure of matter) — why thermal expansion is visible.

INTRODUCED · L01 Kinetic Energy 动能

Energy of motion, $E_k = \frac{1}{2}mv^2$ — **background formula, not introduced in L01**. Thermal role: temperature reflects the *average* KE of particles — in the brick, of atoms vibrating about their sites.

INTRODUCED · L01 Thermodynamics 热力学

The study of how macroscopic variables summarize $\sim 10^{23}$ particles and the rules those summaries obey.

INTRODUCED · L01 Heat 热

Energy in transit driven by ΔT . Carriers: molecular collisions; radiation (IR). ⚠ An object contains internal energy — it does **not** contain heat. L02 grounds heat flow in conservation of energy.

INTRODUCED · L01 Thermal Equilibrium 热平衡

Endpoint of heat flow: common temperature; **net** exchange = 0 while two-way collisions continue. L02 adds: equilibrium also requires temperature to be **uniform inside** each object (slide 15).

PREVIEWED · L01 Phase Change 相变

Solid/liquid/gas transitions. L01: vapour → frost on the -78.5°C brick (deposition, 凝华; slides' shorthand "condensation"). Latent heat comes later.

PREVIEWED · L01 Thermal Expansion 热膨胀

L02: every linear dimension grows with temperature — typical behaviour, with exceptions. Quantitative law next.

PREVIEWED · L01 Oscillations 振动

Repeated motion about an equilibrium position — already true inside the brick. Second half builds single oscillators → waves.

PREVIEWED · L01 Waves 波

Travelling **collective motion**: energy handed neighbour to neighbour without the material travelling. Wave equation later.

INTRODUCED · L01 Electromagnetic Radiation 电磁辐射

Every object emits at any temperature; wavelength set by temperature; room-T → infrared (invisible). L02 adds the spectrum map: gamma ($\sim 10^{-14}$ m) → AM ($\sim 10^4$ m); visible 400–700 nm; hotter → more IR.

INTRODUCED · L02 Conservation of Energy 能量守恒

Total energy of an **isolated system** is constant; energy transfers between parts and changes forms; thermo tracks microscopic KE + PE. **In L02 Clicker 1 the instructor keyed Emmy Noether** (conservation derived from time symmetry) and used **James Joule** — the energy unit — as the trap.

INTRODUCED · L02 Zeroth Law of Thermodynamics 热力学第零定律

A–B and A–C in equilibrium $\Rightarrow$ B–C in equilibrium. "Otherwise, temperature wouldn't make sense!" It is what makes thermometers legitimate. Named "zeroth" because it was recognized after the first and second laws already had numbers.